Solution of Problems on Coulomb's Law –

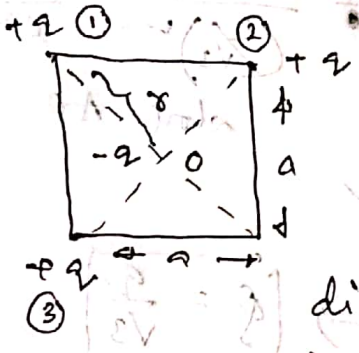
(1) $F = K\frac{q_1q_2}{r^2} = K\frac{q_1q_2}{(1)^2} = \frac{q_1q_2}{4\pi\epsilon_0}$

$F = F_{min}$ if q_1 & q_2 = min^m = e

= electronic charge

$\therefore F_{min} = \frac{e\times e}{4\pi\epsilon_0} = \frac{e^2}{4\pi\epsilon_0}$

(2)



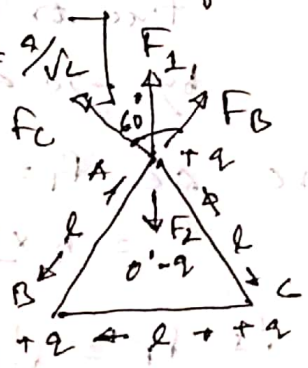
Force on $-q$ at O due to $+q$ at ② & $+q$ at ③ are equal in magnitude & opposite in dir^n. So their resultant is 0.

Hence Force on '$-q$' at O due to $+q$ at ①

is $F = K\frac{q^2}{r^2} = \frac{1}{4\pi\epsilon_0}\frac{q^2}{(a/\sqrt{2})^2} = \frac{q^2}{2\pi\epsilon_0 a^2}$

$\left[r = \frac{a\sqrt{2}}{2} = a/\sqrt{2}\right]$

(3) Case – 1



Force on charge $+q$ at A due to $+q$ at B & C are $F_B = F_C = \frac{Kq^2}{l^2} = F$ (say)

Their resultant is $F_1 = \sqrt{F^2+F^2+2F.F\cos 60^\circ}$

$= \sqrt{3}F = \sqrt{3}\frac{Kq^2}{l^2}$ along OA.

Now force on $+q$ at A due to $-q$ at O

is $F_2 = \frac{Kq^2}{(AO)^2} = \frac{Kq^2}{(l/\sqrt{3})^2} = \frac{3Kq^2}{l^2}$ along AO

$\langle AO = l/\sqrt{3}\rangle$

$\therefore F_2 > F_1$ So charge $+q$ at A will move towards centre O.

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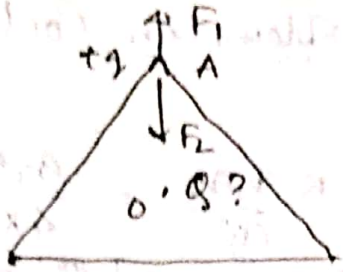
Case - 2



Let for equilibrium of charge +q at A charge Q be placed at centre.

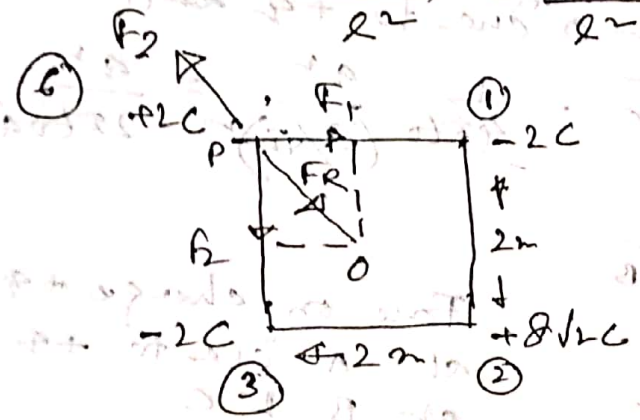
Then F_2 should balance F_1, Hence Q must be '-ve'.

$F_1 = \frac{\sqrt{3}Kq^2}{l^2}$ along AA, $F_2 = \frac{KqQ}{(l/\sqrt{3})^2} = \frac{3KqQ}{l^2}$ along AO.

For equilibrium of q at A,

$F_2 = F_1$

$\frac{3KqQ}{l^2} = \frac{\sqrt{3}Kq^2}{l^2} \Rightarrow \boxed{Q = \frac{q}{\sqrt{3}}}$



Force on +2C at P by -2C at 1 & 3 are $F_1 = F_2$

$= \frac{K \times 2 \times 2}{2^2} = KN$

Their resulter is $F_R = \sqrt{F_1^2 + F_2^2}$ $\because F_1 \perp F_2$

$= \sqrt{K^2 + K^2}$

$= K\sqrt{2}N$ along $\rightarrow$ PO.

Now force on +2C at P by charge $+8\sqrt{2}C$ is

$F_2 = \frac{K(8\sqrt{2})(2)}{(2\sqrt{2})^2}$

$= \frac{16\sqrt{2}K}{8} = 2\sqrt{2}K$ along OP.

$\because F_2 > F_1$

$\therefore$ Net force on +2C at P

$= F_2 - F_1 = 2\sqrt{2}K - \sqrt{2}K$

$= \sqrt{2}K$ along OP.

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(7) Let charges on two bodies be $+q$ & $(Q-q)$ ∴ force betn them:

$$F = \frac{K\, q\,(Q-q)}{r^2} \quad [r = \text{separation}]$$

$$\because \frac{K}{r^2} = \text{const} = C$$

$$\therefore F = C(qQ - q^2) \quad \therefore F = f(q)$$

Condn for max^m value of F is:

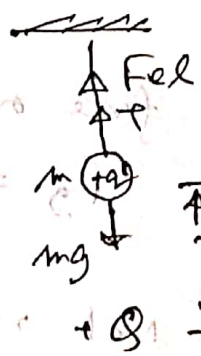
$$\frac{dF}{dq} = 0$$

$$C(Q - 2q) = 0 \quad \therefore 2q = Q \quad \boxed{q = \frac{Q}{2}}$$

$$\therefore \boxed{Q-q} = Q - \frac{Q}{2} = \boxed{\frac{Q}{2}}$$

∴ Charge Q must be divided equally among two bodies.

(9) Let charge on bob be $+q$, for its equilibrium



$$T + F_{el} = mg$$

(Tension in string) (electric force on +q by +Q) (force of gravity)

If string is loose, $T = 0$

Then $F_{el} = mg$

$$K\frac{qQ}{r^2} = mg \rightarrow \boxed{q = \frac{mgr^2}{KQ}}$$

$$K = \frac{1}{4\pi\epsilon_0}$$

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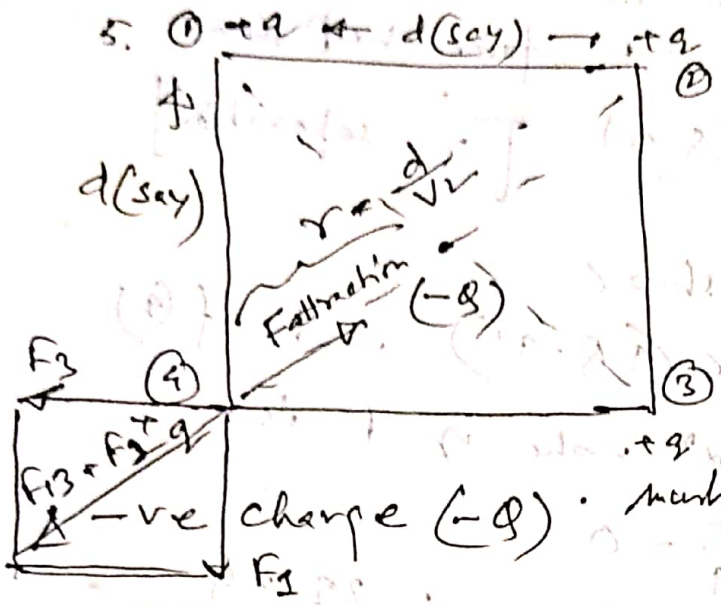
5.



Since the charge distribution is symmetrical, so for equilibrium of all charges at vertices, a −ve charge (−Q) must be placed at centre

Forces on any charge +q at any vertex say that at ④, by charge +q at ① & ③ are $F_1 = F_3 = K\frac{q^2}{d^2} = F$ (say)

$\because F_1 \perp F_3$, their resultant $F_{13} = \sqrt{F_1^2 + F_3^2}$

$$F_{13} = \sqrt{F^2 + F^2} = F\sqrt{2}$$

$$= \frac{Kq^2\sqrt{2}}{d^2}$$

Now force on +q at ④ due to +q at ②

is $F_2 = \frac{Kq^2}{(d\sqrt{2})^2} = \frac{Kq^2}{2d^2}$

So Net repulsion on charge +q at ④ due to charges +q, +q, +q at corners ①, ②, ③

is $F_{repulsion} = F_{13} + F_2 = \frac{Kq^2\sqrt{2}}{d^2} + \frac{Kq^2}{2d^2}$

$$= \frac{Kq^2}{2d^2}(1 + 2\sqrt{2}), \text{ diagonally outward.}$$

For equilibrium of +q at ④, Force of attraction on it by −Q at centre is $F_{attraction} = \frac{KqQ}{\left(\frac{d}{\sqrt{2}}\right)^2}$

$$= \frac{2KqQ}{d^2} \text{ diagonally inward.}$$

if +q at ④ is at rest,

$F_{attraction} = F_{repulsion} \Rightarrow \frac{2KqQ}{d^2} = \frac{Kq^2}{2d^2}(1 + 2\sqrt{2})$

or $\boxed{Q = \frac{q}{4}(1 + 2\sqrt{2})}$

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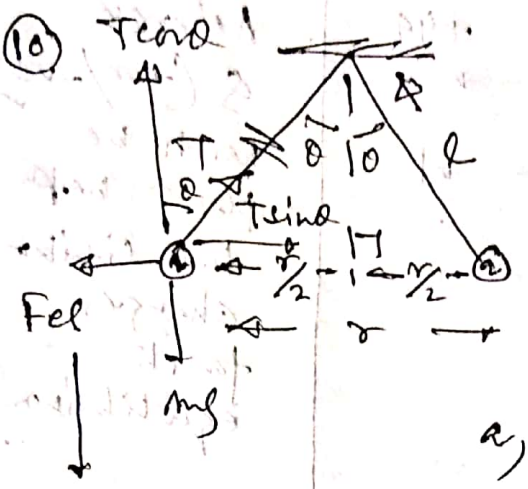
(8) solved in ~~std~~ online class 1.

(10)



$F_{el} = \frac{kq^2}{r^2}$.

For equilibrium of any charge,

$T\sin\theta = \frac{kq^2}{r^2}$ — (1)

$T\cos\theta = mg$ — (2)

(1) ÷ (2)

$\therefore \tan\theta = \frac{kq^2}{mg\,r^2}$

$\because \theta$ = small

$\tan\theta \approx \sin\theta = \frac{r/2}{l}$

$= \frac{r}{2l}$.

$\frac{r}{2l} = \frac{kq^2}{mg\,r^2}$

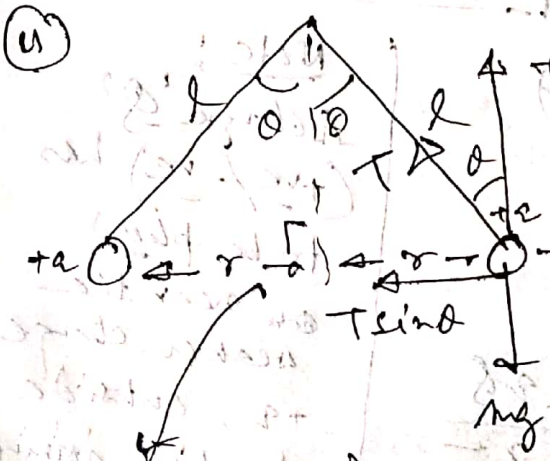
$r^3 = \frac{2kq^2 l}{mg}$

$r = \left(\frac{2kq^2 l}{mg}\right)^{1/3}$

Ref: if θ is small
$\sin\theta \approx \theta$
$\cos\theta \approx 1$
$\therefore \tan\theta \approx \theta \approx \sin\theta$

— o —

(11)



FBD of any q

$F_{el} = \frac{kq^2}{(2r)^2} = \frac{kq^2}{(2l\sin\theta)^2}$

$\because$ each charge is at rest.

$\sin\theta = r/l$
$r = l\sin\theta$
$\therefore 2r = 2l\sin\theta$

$\therefore T\sin\theta = \frac{kq^2}{4l^2\sin^2\theta}$ — (1)

$T\cos\theta = mg$ — (2)

(1) ÷ (2) $\frac{\sin\theta}{\cos\theta} = \frac{kq^2}{(mg)(4l^2\sin^2\theta)}$

$\therefore m = \frac{kq^2\cot\theta}{4gl^2\sin^2\theta} = \frac{q^2\cot\theta}{16\pi\epsilon_0 g l^2\sin^2\theta}$

$\left(\because k = \frac{1}{4\pi\epsilon_0}\right)$

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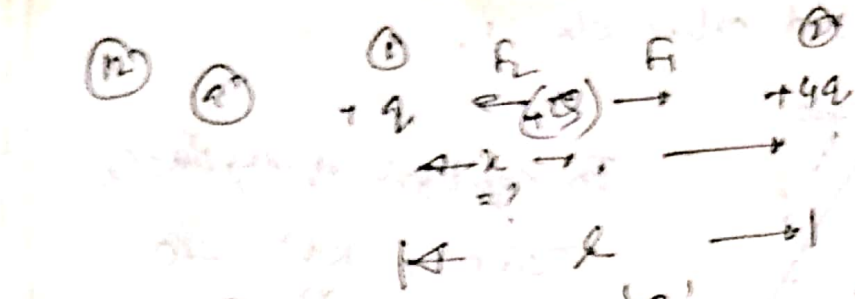
(12) (a)



Note, A charge Q (+ve/−ve) has to be placed betn two similar charges for its equilibrium.

For equilibrium of 'Q'

$$F_1 = F_2$$

$$K\frac{4Q}{x^2} = \frac{KQ(4q)}{(l-x)^2}$$

$$\frac{l-x}{x} = \pm 2$$

for +ve root $\frac{l-x}{x} = +2 \Rightarrow l-x = 2x$

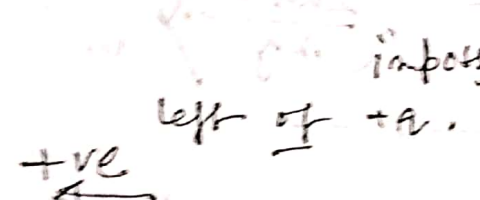
$3x = l$, $\boxed{x = l/3}$

for −ve root, $\frac{l-x}{x} = -2 \Rightarrow l-x = -2x$

$x = -l$

impossible as Q lies to the left of +q. $\therefore \boxed{x = l/3}$

(b)



Note; A charge 'Q' (+ve/−ve) has to be placed on near the weaker charge +q, outside the line joining (+q) & (−9q) for equilibrium of 'Q'

at equilibrium of Q

$$F_2 = F_1$$

$$\frac{K(9q)Q}{(l+x)^2} = \frac{KqQ}{x^2}$$

$$\frac{l+x}{x} = \pm 3$$

for +ve root $l+x = 3x$

$\boxed{x = l/2}$

for −ve root, $\frac{l+x}{x} = -3 \Rightarrow l+x = -3x$

$x = -\frac{l}{4}$ impossible

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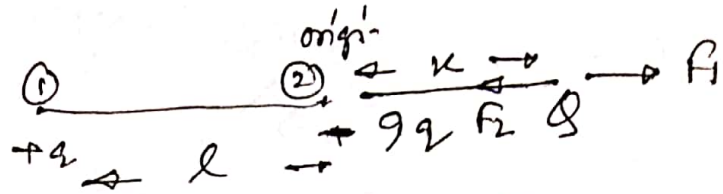
explanation.



If any charge ($+Q$ or $-Q$) is placed betn two opposite charges $+q$ & $-qq$ then Forces on $+Q$ or $-Q$ will act in the same dirn. So equilibrium of any charge ($+Q$ or $-Q$) is not possible betn $+q$ & $-qq$.

if a charge Q is placed nearer stronger charge $-qq$ at a distance x to the right of it the force on Q by $-qq$ i.e $F_2 = \frac{KqqQ}{x^2}$ is very much greater than force on Q by $+q$ i.e

$F_1 = \frac{KqQ}{(l+x)^2}$ i.e $\underline{F_2 >> F_1}$

($\because$ numerator of F_2 is greater than that of F_1 & denominator of F_2 is smaller than that of F_1)

So Q has to be placed to the left & closer to weaker charge q so that its electrostatic force can balance the electrostatic force of the farther stronger charge $-qq$.

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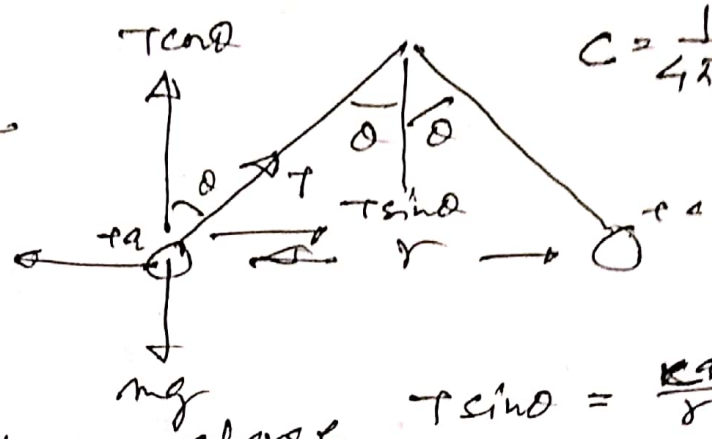
in air

$C = \frac{1}{4\pi\epsilon_0}$



$F_{el} = \frac{kq^2}{r^2}$

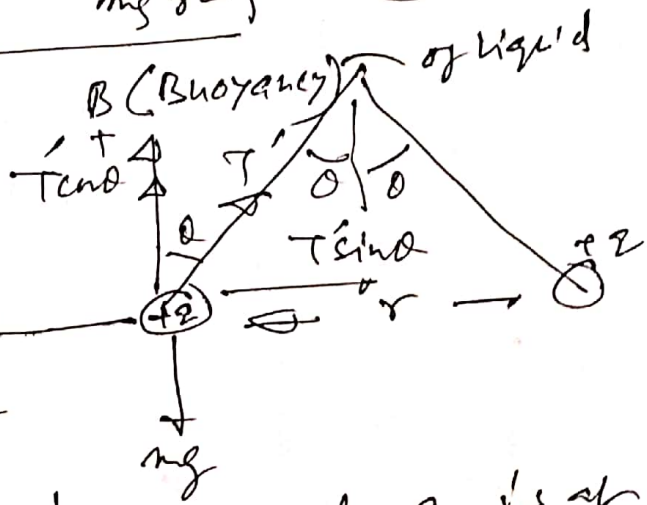
for rest of any charge

$T\sin\theta = \frac{kq^2}{r^2}$ — (1)

$T\cos\theta = mg$ — (2)

(1) ÷ (2) $\boxed{\tan\theta = \frac{kq^2}{mgr^2}}$ — (A)

in Liquid



$F_{el} = \frac{Cq^2}{Kr^2}$

K = dielectric const of liquid

$\left[F_{medium} = \frac{F_{air}}{K}\right]$

∵ each q is at rest.

∴ $T'\sin\theta = \frac{Cq^2}{Kr^2}$ — (1)

$T'\cos\theta + B = mg$

$T'\cos\theta = mg - B$ — (2)

(1) ÷ (2) $\boxed{\tan\theta = \frac{Cq^2}{K(mg-B)r^2}}$ — (B)

from (A) & (B) $\frac{Cq^2}{mgr^2} = \frac{Cq^2}{K(mg-B)r^2}$

$K = \frac{mg}{mg-B}$

mg = vol of bob × density of bob × g = $V l_b g$

$K = \frac{V l_b g}{V l_b g - V l_a g} = \frac{l_b}{l_b - l_a} = \frac{1.6}{1.6-0.8} = 2$

B = Buoyancy = vol of bob × density of liquid × g = $V \times l_a \times g$

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